Recall projective cover last week. if exist then unique.

Corollary: Let P, Q be finite generated projective modules

Then $P \simeq Q \iff P/\mathrm{Rad}\,P \simeq Q/\mathrm{Rad}\,Q$

proof: Consider $\pi: P \to P/\mathrm{Rad}\,P$ $\pi': Q \to Q/\mathrm{Rad}\,Q$

π, π' are essential by Nakayama lemma.

Hence P is the projective cover of $P/\mathrm{Rad}\,P$, as well as Q.

$\Rightarrow$ Obvious

$\Leftarrow$ By the uniqueness of projective cover.

Definition: Let R be a unital ring. We call R is a left Noetherian ring if ${}_R R$ is Noetherian as a left R-module.

Remark: Similar definition for right.

Similar definition for Artinian.

A left-Notherian/Artinian ring may not a right-Notherian/Artinian ring

Theorem: (Levitzki) left Artinian rings are left Noetherian rings.

Theorem: If R is a left Noetherian / Artinian ring. Then every finite generated left R-module is left Noetherian / Artinian.

proof: Let M be a finite generated R-module

$M = Rm_1 + \cdots + Rm_n$

Rm_i is a homomorphic image of ${}_RR$

Hence Rm_i is Noetherian.

M is a finite sum of Noetherian modules

Therefore M is Noetherian.

Theorem: Let ${}_RV = V_1 \oplus \cdots \oplus V_n$ be a direct sum of R-modules. Suppose that $V_1, \ldots, V_n$ are isomorphic to each other.

Then $End_R(V)$ is isomorphic to the full matrix ring of degree n over $End_R(V_1)$

proof: Let $\theta_i : V_1 \xrightarrow{\sim} V_i$ be an isomorphism

Let $\pi_i : V \longrightarrow V_i$ be the projection

For $\sigma \in End_R(V)$, Let $\varphi_{ij}(\sigma) : \theta_i^{-1} \pi_i \sigma \theta_j : V_1 \rightarrow V_1$

$\varphi_{ij}(\sigma) \in E_1 : End_R(V_1)$

Define a map $\varphi: \mathrm{End}_R(V) \longrightarrow M_n(E_1)$

$\sigma \longmapsto (\varphi_{ij}(\sigma))$

1° check this is a ring homomorphism

2° check this is an isomorphism.

- if $\varphi(\sigma)=0$. then $\pi_i \sigma(V_j)=0 \quad \forall\, i,j$

so $\sigma(V_j)=0 \Rightarrow \sigma=0 \Rightarrow$ injective

- let $(\sigma_{ij}) \in M_n(E_1)$, let $\tau_i: V_i \hookrightarrow V$

be the inclusion.

Set $\sigma = \sum_{i,j} \tau_i \theta_i \sigma_{ij} \theta_j^{-1} \pi_j \in \mathrm{End}_R(V)$

Only need to check $\varphi(\sigma) = (\sigma_{ij})_{n\times n}$

Definition: A ring is called semisimple if ${}_RR$ is semisimple as a left R-module

lemma: The followings are equivalent for a unital ring R:

(1). R is a semisimple ring

(2). Every R-module is semisimple

proof: (1) $\Rightarrow$ (2) ${}_RR = \sum_\lambda I_\lambda$ a sum of simple R-modules.

Then $V = \sum_{v\in V} (\sum_\lambda I_\lambda) v \qquad f: I_\lambda \longrightarrow I_\lambda v$

$i \longmapsto iv$

$I_a V = \begin{cases} I_a \\ 0 \end{cases}$ since I_r is simple

(2) $\Rightarrow$ (1) obvious

Lemma: Let $_RM$ be an R-module. $M \simeq \mathrm{Hom}_R({}_RR_R, {}_RM)$

$$\mathrm{Hom}_R(R, M) \longrightarrow M$$
$$f \longmapsto f(1)$$
$$(r \mapsto rm) \longleftarrow\!\shortmid\; m$$

Corollary: $\mathrm{End}({}_RR) \simeq R^{op}$

$$f \longmapsto f(1)$$

pmf: $(f_1 f_2)(1) = f_1(f_2(1)) = f_2(1) f_1(1)$

lemma: $M_n(R)^{op} \simeq M_n(R^{op})$

$$A \longmapsto A^T$$

Theorem (Artin-Wedderburn)

Let A be a semisimple Artinian ring. Then A is a direct sum of matrix rings over division rings.

In particular, if $_AA \simeq S_1^{n_1} \oplus \cdots S_r^{n_r}$ where S_i's are non-isomorphic simple modules occuring with multiples $n_1, \ldots, n_r$. Then $A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$

where $D_i = \mathrm{End}_A(S_i)^{op}$

proof: Recall. $\mathrm{Hom}(M_1 \oplus M_2, N) \simeq \mathrm{Hom}(M_1, N) \oplus \mathrm{Hom}(M_2, N)$

$$\mathrm{Hom}(M, N_1 \oplus N_2) \simeq \mathrm{Hom}(M, N_1) \oplus \mathrm{Hom}(M, N_2)$$

$$\mathrm{End}_A({}_AA) \simeq \mathrm{End}_A(S_1^{n_1}) \oplus \cdots \oplus \mathrm{End}_A(S_r^{n_r})$$

since if $i \neq j$ then $\mathrm{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0$

By schur's lemma $\Rightarrow D_i := \mathrm{End}_A(S_i)$ are division rings

By previous result. $\mathrm{End}_A(S_i^{n_i}) \simeq M_{n_i}(\mathrm{End}_A(S_i))$

$$\simeq M_{n_i}(D_i^{op}) \simeq M_{n_i}(D_i)^{op}$$

And $A \simeq \mathrm{End}_A(A)^{op} \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$

Recall the definition of simple ring, it shows they are matrix rings over a division ring.

We will prove it later.

Definition: Let R be a commutative ring with 1, A be a ring with 1. (associative)

We call A an R-algebra if A is an R-module such that

$a(xy) = (ax)y = x(ay)$, $\forall a \in R$, $x, y \in A$.

Equivalently, there is a ring homomorphism $R \to A$ such that the image of R lies in the center of A. If moreover, R is a field. A is finite dimensional over R. Then we say the algebra A is finite dimensional.

For a field k, a finite dimensional k-algebra A is both Noetherian and Artinian.

Example: R commutative ring with 1, the followings are R-algebras:

(1). $R[x_1, \dots, x_n]$

(2). $M_n(R)$

(3). Let G be a finite group. The group algebra

$$RG = \left\{ \sum_{g \in G} a_g g \mid g \in G, a_g \in R \right\}$$

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{\ell \in G} \sum_{\ell = gh} a_g b_h gh$$

Lemma (Schur, for k-algebra)

Let A be a k-algebra, where k is a field. Let S be a simple A-module, then $\mathrm{End}_A(S)$ is a division ring. Which is a k-algebra.

If A is finite dimensional. $End_A(S)$ is finite dimensional.

Furthermore. if A is finite dimensional and k is algebraically closed. Then $End_A(S) \simeq k$

proof: 1° $End_A(S)$ is k-algebra.

$\forall\, a \in k$. $f \in End_A(S)$, $x \in S$

$(af)(x) = a\, f(x)$

2°. S is a A-module. exist

$$k \xrightarrow{\varphi} A \xrightarrow{\psi} End_A(S)$$

Every A-module is a k-module hence S is a k space.

Consider $End_k(S)$

and $End_A(S) \subseteq End_k(S)$

Let M be a R-module

$End(M) = \{ f : M \to M$. Abel group endomorphisms$\}$

$\Rightarrow \exists$ $R \to End(M)$ $r \mapsto \varphi_r$

$End_R(M) = \{ f : M \to M : R$- homomorphism$\}$

it is $\begin{cases} \text{abel group homomorphism} & f(x+y) = f(x) + f(y) \\ \text{commute with } R\text{-action} & f(rx) = rf(x) \end{cases}$ $+$

$\Rightarrow \forall r \in R, \ \exists \ \varphi_r : M \to M \ \in \mathrm{End}(M)$

$m \mapsto rm$

$\Rightarrow f$ commute with φ_r, $\forall r \in R$.

$\Rightarrow \ \mathrm{End}_R(M) = \{ f \in \mathrm{End}(M) \mid f \text{ commutes with } \varphi_r, \forall r \in R \}$

Now A an R-algebra, M an A-module

$\mathrm{End}_A(M) = \{ f \in \mathrm{End}_R(M) \mid f \text{ commutes with (action of) } A \}$

If A f.d. then $\mathrm{End}_A(S)$ is k-subspace of $\mathrm{End}_k(S)$

*. S is f.d over k since S simple and A f.d

since $S \cong A / \mathrm{Ann}_A(x)$ $x \in S$, $\langle x \rangle = S$

$\Rightarrow S$ f.d $\Rightarrow \mathrm{End}_k(S)$ fin. dim $\Rightarrow \mathrm{End}_A(S)$ fin. dim.

If k is algebraically closed.

$f \in \mathrm{End}_A(S) \subseteq \mathrm{End}_k(S)$

since $k = \bar{k}$, f has an eigenvalue λ

Consider $f - \lambda I$, this is not invertible in $\mathrm{End}_k(S)$

hence $\ker(f - \lambda I) \neq 0$

and $f - \lambda I$ is furthermore in $\mathrm{End}_A(S)$.

By Schur's lemma, $\ker(f - \lambda I) = S \Rightarrow f - \lambda I = 0$

Hence $\operatorname{End}_A(S) = \{\lambda I \mid \lambda \in k\} \simeq k$

Theorem: Let $k = \bar{k}$. A finite dimensional semisimple k-alg A is isomorphic to a direct sum of matrix algebras over k

that is $A \simeq M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k)$

Corollary: Let A be a finite dimensional semisimple k-algebra. k a field

In any decomposition. ${}_AA = S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where

S_i's are non-isomorphic simple A-modules

we have $S_1, \dots, S_r$ is a complete set of representations of ~~the~~ isomorphism classes of simple A-modules.

If moreover $k = \bar{k}$, then $n_i = \dim_k S_i$ and

$\dim_k A = n_1^2 + \cdots + n_r^2$

proof: S is simple, $S \simeq A/M$ where M is a maximal submodule.

By semisimple $\Rightarrow$ $A = M \oplus S$

Hence all simple submodule appear in $\{S_1, \dots, S_r\}$.

Since $k=\bar{k}$ we have

$$A \simeq M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k)$$

$$M_{n_i}(k) = \underbrace{\left\{ \begin{pmatrix} * & 0 & \cdots & 0 \\ \vdots & & & \\ * & 0 & \cdots & 0 \end{pmatrix} \right\} \oplus \cdots \oplus \left\{ \begin{pmatrix} 0 & \cdots & 0 & * \\ & & & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix} \right\}}$$

each of them is simple $M_{n_i}(k)$ module.

Definition: A (non-zero) R-module M is called indecomposable if $M = M_1 \oplus M_2 \Rightarrow M_1 = 0$ or $M_2 = 0$

Theorem: (Krull-Schmidt) Let M has a composition series. Then

$M = V_1 \oplus \cdots \oplus V_m$

$M = V_1' \oplus \cdots \oplus V_n'$

where V_i, V_i' indecomposable

implies $m=n$ and $V_i \simeq V_i'$ after reordering

Idempotents: Let R be a ring with 1_R

An idempotent of R is a non-zero element $e \in R$ with $e^2 = e$

Fact: • $eRe = \{ere \mid r \in R\}$ is a ring with identity e

- Two idempotent $e_1, e_2 \in R$ are called orthogonal if $e_1 e_2 = e_2 e_1 = 0$

- n idempotents are called orthogonal if any two of them are orthogonal

- If $e_1 \dots e_n$ are orthogonal idempotents of R, then $e_1 + \dots + e_n$ is an idempotent

- An idempotent decomposition of an idempotent $e \in R$ is $e = e_1 + \dots + e_n$ where $e_1, \dots, e_n$ are orthogonal idemporents.

- If $e = e_1 + e_2$ is an idempotent decomposition then $e_1, e_2 \in eRe$ (consider $e e_1 e = e_1$)

- An idempotent is called primitive if it is not a sum of two orthogonal idempotents.

A idempotent decomposition $e = e_1 + \dots + e_n$ is called a primitive orthogonal decomposition if $e_1 \dots e_n$ are primitive.

Example: let e be an idempotent.

then $e(1-e) = 0$ $(1-e)^2 = 1-e$ is and idempotent.

hence $1 = e + (1-e)$ is an idempotent decompotition of 1.

• Let e be an idempotent

$Re = \{ae \mid a \in R\}$ is a left ideal of R

Theorem: Let e be an idempotent of R

If $e = e_1 + \cdots + e_n$ is an idempotent decomposition

$$Re = Re_1 \oplus \cdots \oplus Re_n$$

conversly if Re is a direct sum of R-ideal

$$Re = I_1 \oplus \cdots \oplus I_n$$

then there exist an idempotent decomposition

$e = e_1 + \cdots + e_n$ with $I_i = Re_i$

In this way the idempotent decomposition of e

$\xleftrightarrow{1:1}$ derect sum decomposition

of ${}_R(Re)$

proof: If $e = e_1 + \cdots + e_n$ idempotent decomposition

$e_i e = e e_i = e_i$ hence $Re_i \subseteq Re$

$(\forall x \in Re,\ xe = x)$

$\forall a \in R$, $ae = ae_1 + \cdots + ae_n \in Re_1 + \cdots + Re_n \Rightarrow Re = Re_1 + \cdots + Re_n$

Let $ae = a_1e_1 + \cdots + a_ne_n$

$\Rightarrow ae_i = aee_i = a_ie_i \Rightarrow$ expression is unique.

$Re = Re_1 \oplus \cdots \oplus Re_n$

Conversely Let $Re = I_1 \oplus \cdots \oplus I_n$ exist a decomposition.

$e = e_1 + \cdots + e_n$, $e_i \in I_i$

for any $a_i \in I_i \subseteq Re$

$a_i = a_i e = a_i e_1 + \cdots + a_i e_n$

$= a_i e_i$ and $a_i e_j = 0$ if $j \neq i$

$I_i = Re_i$

let $a_i = e_i$, $e_i^2 = e_i$, $e_i e_j = 0$ if $j \neq i$

$\Rightarrow$ $e = e_1 + \cdots + e_n$ is an idempotent decomposition.

Corollary: $e \in R$ an idempotent. the followings are equivalent

(1). e is primitive

(2) $_R(Re)$ is indecomposable as a left R-module

(3) e is the unique idempotent of eRe